

Anchoring Options to an Incomplete Forward Curve: Identification and Out-of-Sample Evidence from Turkish Electricity

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Abstract

Electricity options in emerging markets must be valued without the two ingredients that standard derivative pricing takes for granted: a traded option surface from which risk-neutral parameters could be read off, and a continuous forward curve. The Turkish market offers only a sparse strip of monthly baseload forward quotes, and even that strip leaves the nearest delivery month unquoted. This paper develops a valuation framework that separates what such a market identifies from what it does not. An hourly curve is constructed as the smoothest one consistent with the observed monthly averages, imposed as hard equality constraints, and a two-regime Ornstein–Uhlenbeck residual with residual-demand-driven time-varying transition probabilities is attached in a centred form, so that $\mathbb{E}^{\mathbb{Q}}[P_t] = F(t)$ holds for every admissible parameter vector. The construction isolates the forward level as the only market-identified object, leaving volatility and regime risk premia as explicit, separately reported assumptions rather than implicit consequences of a fitting procedure. Expiry-hour European options are priced on this structure through a coupled moment-ODE and finite-difference system, cross-validated against Monte Carlo simulation, with the full sensitivity of the resulting prices to each unidentified input reported. The framework is then evaluated against realised prices. Over the first half of 2026 an exceptional hydrological year drove Turkish spot prices far below the December 2025 forward strip, producing a mean bias of $-1,019$ TRY/MWh across

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5,088 delivery hours. Replaying the curve construction at seven valuation dates between 2022 and 2025 shows that this outcome was not exceptional: the pooled bias over 49 date-month observations averages -603 TRY/MWh (-18%), realised prices fell below the strip at five of seven dates and rose above it at two, and the 2026 result lies inside the interquartile range of the date means. Since the model reproduces the strip exactly, this error is inherited from the forward market rather than generated by the pricing layer. The monthly strip is therefore a biased and sign-unstable predictor of realised baseload prices, and that, rather than any property of the pricing scheme, bounds what forward-anchored valuation can deliver in this market.

Keywords: electricity option valuation, regime switching, time-varying transition probabilities, hourly forward curve, Turkish electricity market, incomplete markets

1. Introduction

Valuing an electricity option requires a probability measure under which the expected discounted payoff is the price. Where a liquid option surface exists, that measure is recovered from traded instruments: implied volatilities pin down the diffusion and the forward curve pins down the drift. Neither ingredient is available in most emerging electricity markets. The Turkish market is a representative case. Its organised derivative segment, the *Vadeli Elektrik Piyasası* (VEP) operated by EPIAŞ, lists monthly baseload futures, and exchange-traded options on power are effectively absent. A practitioner who needs the value of a call on the hourly spot price at a specific delivery hour therefore faces two gaps at once: the forward curve is observed only as a coarse strip of monthly averages, and the risk-neutral distribution around that curve is not observed at all.

Each gap has been addressed separately in a mature literature. On the distributional side, regime switching in power prices is well established. Building on Hamilton (1989), Huisman and Mahieu (2003) and Deng (2000) argue that spike behaviour is better represented by a switch between a calm mean-reverting state and a distinct spike state than by adding jumps to a single process. Janczura and Weron (2010) compare specifications systematically and find that independent-spike-regime models with heavy tails dominate simpler alternatives, consistent with the tail evidence in Weron (2009); Deschatre et al. (2021) survey the resulting model class. Allowing

the switching probabilities to respond to fundamentals, following Diebold et al. (1994) and Filardo (1994), tightens the fit further: Mount et al. (2006) and Kanamura and Ohashi (2008) make transition probabilities depend on load and reserve margin, and Karakatsani and Bunn (2008) show that intra-day regime dynamics carry economic content rather than statistical artefact. Residual demand, load net of non-dispatchable renewable generation, is the natural covariate. Wagner (2014) and Do et al. (2021) document its role in price formation, and the merit-order literature (Sensfuß et al., 2008; Ketterer, 2014; Clò et al., 2015) explains the mechanism: renewable output displaces the marginal thermal unit and shifts the supply stack. For Türkiye, Sirin and Yilmaz (2020) quantify the merit-order effect through quantile regression, while Ugurlu et al. (2018) and Ertuğrul et al. (2022) document the forecasting properties of Turkish spot prices.

On the curve-construction side, turning coarse forward quotes into a granular curve is equally well studied. Lucia and Schwartz (2002) establish the decomposition of a power price into a deterministic component and a mean-reverting stochastic residual that underlies most subsequent work; Fleten and Lemming (2003) pose curve construction as a smoothing problem constrained to reproduce observed contract prices; Borak and Weron (2008) model forward curve dynamics semi-parametrically; Kiesel et al. (2019) and Caldana et al. (2017) construct hourly curves for European markets. The common structure is a smoothness objective subject to arbitrage-free reproduction of the observed strip.

Pricing on a regime-switching state space is likewise a solved numerical problem. Buffington and Elliott (2002) and Jobert and Rogers (2006) give the Markov-modulated framework, Boyle and Draviam (2007) and Huang et al. (2011) develop finite-difference schemes for the coupled systems, Chen and Forsyth (2010) apply them to gas storage, and Zeng et al. (2017) compare finite-difference and Monte Carlo approaches directly. The closest antecedent to the present study is Paraschiv et al. (2015), who combine spot and forward information within a regime-shifting specification. Our departure is to treat the boundary between the two as an identification question rather than an estimation one.

That question is what remains open. When a market supplies a forward strip and nothing else, which parameters of such a model are determined by data, and which are assumptions the analyst has imported? The distinction is easily obscured. A model calibrated to reproduce a forward strip to numerical precision looks calibrated. Where switching parameters and

volatilities have been estimated on a historical sample under the physical measure $\mathbb{P}$ and are then used for pricing under $\mathbb{Q}$ without adjustment, the resulting value embeds an untested assumption that the market charges no premium for regime or volatility risk. Janczura (2014) constructs such premia explicitly within a regime-switching framework, Bessembinder and Lemmon (2002) derive equilibrium forward premia from hedging pressure, and Longstaff and Wang (2004) and Algieri et al. (2021) measure them empirically. The premium is a real economic object; it does not vanish because it was not estimated.

Contribution.. The separation is made explicit here, and a valuation framework is built around it. Four contributions follow.

First, a *forward-centred* representation is proposed, in which the spot price is written as $P_t = F(t) + X_t - \mathbb{E}^{\mathbb{Q}}[X_t]$, with F the equality-constrained hourly curve and X a two-regime Ornstein–Uhlenbeck residual with time-varying transition probabilities. Subtracting the residual’s own mean is not a normalisation of convenience: it renders $\mathbb{E}^{\mathbb{Q}}[P_t] = F(t)$ an algebraic identity holding for every admissible parameter vector, so that the market-identified first moment is structurally insulated from the unidentified higher moments. This is established in Proposition 1 and verified numerically across the parameter space explored below, including under measure changes.

Second, the unidentified inputs are made auditable rather than invisible. Every parameter carries a provenance label distinguishing calibrated quantities from inherited historical estimates, derived reconstructions and outright assumptions (Table 2), and each is subjected to a controlled sensitivity experiment in which it alone varies. The risk-neutral drift channel is implemented as an explicit, switchable perturbation whose default value of zero reproduces the physical-measure baseline exactly, which makes the zero-premium assumption a visible choice rather than a silent default.

Third, the framework’s principal weakness is shown to lie outside its pricing layer. Using 5,088 hours of spot prices realised after the valuation date, we find that the December 2025 forward strip missed the subsequent outcome by a mean of $-1,019$ TRY/MWh, with monthly signed errors from -24% to -76% , during a period of exceptionally strong hydrological conditions and record renewable output. Replaying the curve construction at seven valuation dates from 2022 onward places that outcome in context: it is large but not exceptional, and the strip’s error is negative on average yet unstable in sign. To our knowledge the predictive performance of the Turkish monthly

forward strip has not previously been measured over a multi-year sample, and the resulting distribution bounds what forward-anchored valuation can deliver here.

Fourth, a methodological failure and its correction are documented in full. An initial specification inherited its mean-reversion speed from an autoregressive fit to the untransformed price level, producing a within-regime half-life of roughly nineteen years and residual dispersion seven to thirteen times wider than the realised sample exhibits. Diagnosis required separating within-regime persistence from the marginal persistence of the switching process, quantities that differ here by four orders of magnitude. A framework whose sensitivity to a single inherited coefficient exceeds a factor of four is one whose provenance discipline is not optional, and the episode is reported on that basis.

Section 2 describes the market, contract and data; Section 3 develops the hourly curve; Section 4 specifies the residual dynamics and the centring identity; Section 5 derives the moment system and pricing equation; Section 6 sets out parameter provenance; and Section 7 reports calibration quality, the option surface, the sensitivity battery and the realised evaluation.

2. Market, contract and data

2.1. The Turkish wholesale electricity market

Wholesale electricity in Türkiye clears through EPİAŞ, the market operator. The day-ahead market produces an hourly clearing price, the *Piyasa Takas Fiyatı* (PTF), for every delivery hour of the following day (EPİAŞ, 2025). PTF is the reference price for physical settlement and the underlying taken here for option valuation. Prices are quoted in Turkish lira per megawatt-hour and are subject to administratively set ceiling and floor levels revised periodically (Enerji Piyasası Düzenleme Kurumu, 2024).

The organised forward segment, VEP, lists monthly baseload futures. A contract for delivery month m , denoted $\text{EBM}m\text{myy}$, settles against the arithmetic mean of PTF over every delivery hour of that month. This averaging structure is the first source of the identification problem addressed here. A monthly baseload quote constrains the *average* of 720 to 744 hourly forward prices and says nothing about how that average is distributed across hours. Two curves with identical monthly means but different intraday shapes are equally consistent with the same quote, yet imply materially different values for an option written on a single delivery hour.

Two features make the market a useful laboratory for these questions. Renewable penetration has grown quickly, so residual demand fluctuates over a wide range and the merit-order mechanism transmitting it into price is strong (Sirin and Yilmaz, 2020). And the lira has depreciated substantially over the period from which historical parameters must be drawn, so nominal price series mix genuine dynamics with monetary trend. Consequences of the second feature are taken up in Section 9.

2.2. *The contract*

The instrument valued here is a European call on the PTF observed at a single expiry hour T ,

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(P_T - K)^+], \quad (1)$$

where P_T is the hourly clearing price at T , K the strike and r a flat continuously compounded annual rate. This is not the monthly baseload contract that VEP quotes. The distinction is enforced in the implementation: monthly quotes are represented by an object that raises an error if accessed as a point forward, since conflating the two is the shortcut the data structure most invites.

2.3. *Valuation date and samples*

The valuation timestamp is 2025-12-31 20:00 UTC, the final hour of the historical sample, at which the observed spot is 2,917.78 TRY/MWh. Six monthly baseload quotes were observed at that date, for delivery months February through July 2026 (Table 1). January 2026 carries no quote. This is not an avoidable data gap: the nearest delivery month is the one whose option values matter most for short-dated contracts, and it is precisely the month the strip does not price. Section 3.3 treats it explicitly.

Two further samples enter. The regime covariate is a standardised residual demand series z_t at hourly frequency, standardised using the training scaler of the underlying estimation exercise ($N = 61,361$). The realised evaluation uses hourly PTF for January through July 2026, 5,088 hours, retrieved from the EPIAŞ transparency platform after the fact. The evaluation sample is strictly posterior to the valuation timestamp; no observation from it enters calibration, parameter estimation or model selection. This is verified structurally rather than by assertion, through a guard test that fails if any fitting routine is exposed to a timestamp at or beyond the valuation cutoff.

Table 1: Observed VEP monthly baseload quotes at the valuation date (2025-12-31), in TRY/MWh.

Contract	Delivery month	Delivery hours	Quote
EBM0226	February 2026	672	2,900.99
EBM0326	March 2026	744	2,556.31
EBM0426	April 2026	720	2,500.66
EBM0526	May 2026	744	2,506.25
EBM0626	June 2026	720	2,245.33
EBM0726	July 2026	744	3,558.19
January 2026		744	<i>unquoted</i>

The strip carries information worth noting before any modelling. The market priced a pronounced decline from February to June followed by a sharp July rebound of roughly 58% over the June level, a shape consistent with cooling-driven summer peaks and high spring hydro and solar availability. The hourly curve must therefore accommodate a large discontinuity at the June–July boundary originating in the data rather than in the interpolation method, a point verified in Section 7.1.

3. The hourly forward curve

Figure 1 summarises how the components of the framework fit together and, more importantly, where market information enters and where it does not. The upper path carries observed prices into the forward curve. The lower path carries historical estimates into the residual dynamics. The two meet only at the centring step, and the separation is what makes the identification argument of this paper possible.

3.1. The averaging constraint

Let $\mathcal{H}_m$ denote the set of delivery hours of month m and VEP_m the observed baseload quote. Settlement against the arithmetic mean imposes

$$\frac{1}{|\mathcal{H}_m|} \sum_{h \in \mathcal{H}_m} F(h) = \text{VEP}_m, \quad m \in \mathcal{M}, \quad (2)$$

where $\mathcal{M}$ is the set of quoted months. With N hourly unknowns and $|\mathcal{M}| = 6$ constraints, the system is massively underdetermined: in our configuration

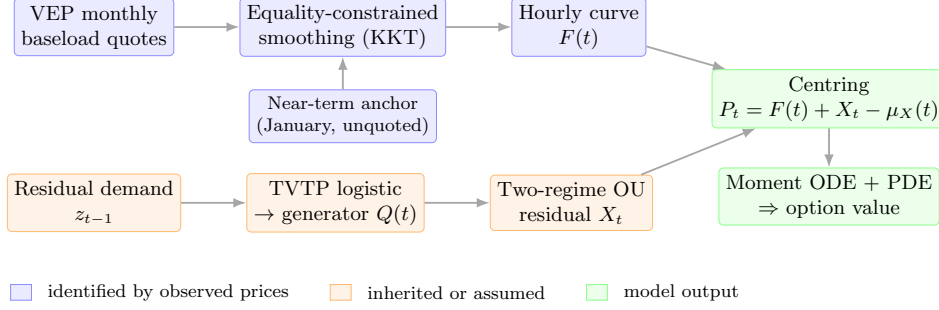


Figure 1: Structure of the valuation framework. Blue elements are identified by observed prices; orange elements are inherited from historical estimation or assumed. Only the first moment of the price distribution is determined by the market path; volatility and regime risk premia are not. (Vector source: `figures/pipeline_src.tex`.)

$N = 5,089$ and the constraints fix six linear functionals. The remaining $N - 6$ degrees of freedom are chosen, not identified. We select them by a smoothness criterion, and we regard that selection as an assumption on the same footing as any other, not as an estimation result.

3.2. Roughness-penalised construction

Write $\mathbf{F} \in \mathbb{R}^N$ for the vector of hourly forward prices and $D_2 \in \mathbb{R}^{(N-2) \times N}$ for the second-difference operator, $(D_2 \mathbf{F})_i = F_{i+2} - 2F_{i+1} + F_i$. Let $\mathbf{B}$ be a piecewise baseline that is constant at VEP_m within each quoted month and follows the near-term anchor rule of Section 3.3 elsewhere. The curve solves

$$\min_{\mathbf{F}} w_R \|D_2 \mathbf{F}\|_2^2 + w_A \|\mathbf{F} - \mathbf{B}\|_2^2 \quad \text{subject to} \quad A\mathbf{F} = \mathbf{b}, \quad (3)$$

where each row of A implements one instance of (2), together with the anchor row. This is a convex quadratic program with linear equality constraints, and we solve it as a single sparse Karush–Kuhn–Tucker system,

$$\begin{bmatrix} 2(w_R D_2^\top D_2 + w_A I) & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} \mathbf{F} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} 2w_A \mathbf{B} \\ \mathbf{b} \end{bmatrix}. \quad (4)$$

Because the quotes enter as equality constraints rather than as penalty terms, the weights w_R and w_A govern only the shape of the curve within months. They cannot trade off fit against smoothness, and no choice of weights can violate (2). This has a consequence for how the resulting fit

should be read, and we state it plainly because the opposite reading is the natural one: the monthly reproduction error we report in Section 7.1 is of order 10^{-12} TRY/MWh, and that number is a property of the solver, not evidence about the model. A formulation that reproduces its inputs by construction has not been tested by reproducing them.

3.3. The unquoted near-term window

January 2026 has no quote, and every option maturity examined here falls inside it. This is not a gap peculiar to one date. Across the seven valuation dates used in Section 7.5, spanning December 2022 to December 2025, the nearest delivery month is unquoted at every one: the front contract is delisted before the month it delivers into begins, so a participant valuing a short-dated instrument never observes a forward price for the period that instrument covers. The level over that window must therefore come from an assumption, and the assumption is a permanent feature of valuation in this market rather than an occasional inconvenience. Three rules are considered, all respecting the observed spot at the valuation hour where applicable:

1. `spot_flat`: hold the observed spot 2,917.78 TRY/MWh flat across the window;
2. `spot_to_next_linear`: interpolate linearly from the observed spot at t_0 to the February quote at the month boundary;
3. `flat_next_month`: hold the February quote flat backwards across January.

We adopt `spot_to_next_linear` as the production rule on three grounds. It satisfies $F(t_0) = P_{t_0}$, so the curve is consistent with the one spot observation that is not in dispute. It joins the February quote without a discontinuity at the month boundary, with a residual jump of 0.05 TRY/MWh against 12–47 TRY/MWh at boundaries elsewhere in the curve. And it produces a non-degenerate term structure across the window, whereas `flat_next_month` violates $\mathbb{E}[P_{t_0}] = P_{t_0}$ by 17 TRY/MWh and `spot_flat` embeds an unmotivated claim that January is a high-price month relative to February. Section 7.3 quantifies how much this choice moves option values, which is the question that matters more than the choice itself.

Remark 1. *The near-term anchor is the weakest link in the market path of Figure 1, and its weakness is structural rather than remediable within the data. Where no quote exists for the delivery period, no construction can*

identify its forward level; it can only be assumed, and the assumption should be reported as a sensitivity range rather than a point.

4. Residual dynamics and the centring identity

4.1. Transform and regime structure

Electricity prices are bounded below by an administrative floor, occasionally negative in other markets (Aust and Horsch, 2020), and heavy-tailed above. The underlying estimation exercise works on the inverse hyperbolic sine transform,

$$y_t = \operatorname{asinh}\left(\frac{P_t}{s_P}\right), \quad P_t = s_P \sinh(y_t), \quad (5)$$

with scale $s_P = 282.48$ TRY/MWh. The transform is odd, defined on the whole real line and asymptotically logarithmic, so it compresses spikes without the domain restriction of a log transform.

Let $S_t \in \{0, 1\}$ be a continuous-time Markov chain, with state 0 the calm regime and state 1 the stress regime, distinguished by their diffusion coefficients $\sigma_0^y \ll \sigma_1^y$. Regime volatilities are inherited from the underlying estimation as $\sigma_0^y = 0.003535$ and $\sigma_1^y = 0.092407$ per $\sqrt{\text{hour}}$, a ratio of roughly 26.

4.2. Time-varying transition probabilities

Transition probabilities depend on standardised residual demand at a one-hour lag through a logistic link,

$$p_{01}(t) = \Lambda(\alpha_{01} + \gamma_{01}z_{t-1}), \quad p_{10}(t) = \Lambda(\alpha_{10} + \gamma_{10}z_{t-1}), \quad \Lambda(x) = \frac{1}{1 + e^{-x}}, \quad (6)$$

following Diebold et al. (1994) and Filardo (1994); consistency of maximum likelihood estimation for this class with covariate-dependent transitions is established by Pouzo et al. (2022), and Bazzi et al. (2017) develop the score-driven alternative. The lag structure matters for interpretation: z_{t-1} is the residual demand observed *before* the transition it governs, so the specification is predictive rather than contemporaneous.

Figure 2 shows the two fitted transition functions against the sample distribution of the covariate. Over the bulk of the sample the calm-to-stress probability exceeds the reverse, and the two cross near $z \approx 1.4$; the economic

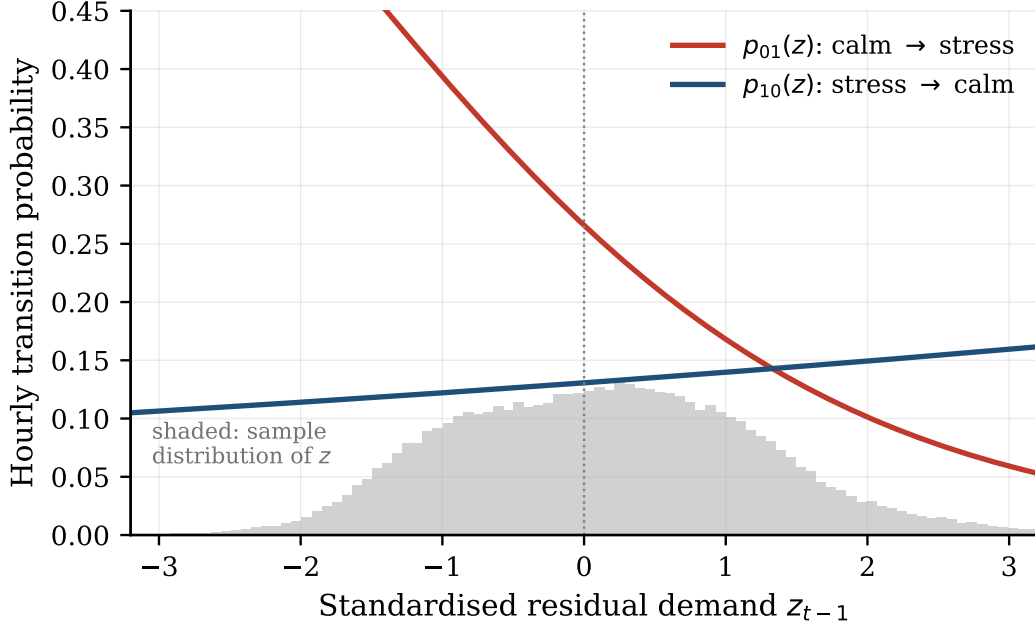


Figure 2: Fitted hourly transition probabilities against standardised residual demand, with the sample distribution of the covariate shaded.

reading of that crossing is deferred to Section 7.3, where the covariate channel is quantified.

The discrete one-step probabilities (6) are converted to continuous-time intensities by matrix logarithm. For the two-state chain with one-step matrix $\Pi(t) = \begin{pmatrix} 1-p_{01} & p_{01} \\ p_{10} & 1-p_{10} \end{pmatrix}$ and $s = p_{01} + p_{10}$, the generator is

$$Q(t) = \begin{bmatrix} -q_{01}(t) & q_{01}(t) \\ q_{10}(t) & -q_{10}(t) \end{bmatrix}, \quad q_{01}(t) = \frac{\lambda(t) p_{01}(t)}{s(t)}, \quad \lambda(t) = -\ln(1 - s(t)), \quad (7)$$

with q_{10} defined symmetrically, reducing to $q_{ij} \approx p_{ij}/\Delta t$ when s is small. The embedding is exact whenever $s < 1$, which holds throughout our sample. Appendix Appendix B documents the regime label reconciliation this conversion required.

4.3. Residual process and price-space diffusion

Conditional on the regime, the residual follows an Ornstein–Uhlenbeck process in price space,

$$dX_t = \kappa (m_{S_t} - X_t) dt + \sigma_{S_t}^X(t) dW_t, \quad (8)$$

with a common mean-reversion rate κ and regime-dependent targets m_i . The diffusion coefficient is mapped from transform space to price space by the delta method applied to (5). Since $dP/dy = \sqrt{P^2 + s_P^2}$, evaluating at the forward level gives

$$\sigma_i^X(t) = \sigma_i^y \sqrt{F(t)^2 + s_P^2}, \quad (9)$$

which is state-dependent through F and therefore time-varying even for fixed σ_i^y . At $F \approx 2,900$ TRY/MWh the factor is approximately 2,914, so the stress-regime price-space diffusion is on the order of 269 TRY/MWh per $\sqrt{\text{hour}}$.

Two properties of (8) deserve emphasis. It is additive rather than multiplicative, so the model can in principle generate negative terminal prices; we report the simulated probability of that event in Section 7 and treat it as a diagnostic of specification adequacy. And κ is shared across regimes, so regime switching enters the second moment but not the speed of mean reversion, which keeps the moment system linear.

4.4. Forward centring

The spot price is defined as

$$\boxed{P_t = F(t) + X_t - \mu_X(t), \quad \mu_X(t) = \mathbb{E}^{\mathbb{Q}}[X_t].} \quad (10)$$

Proposition 1 (Exact first-moment anchoring). *Let X satisfy (8) with regime process generated by (7), let $\mu_X(t) = \mathbb{E}^{\mathbb{Q}}[X_t]$ be computed under the same measure and the same parameters used in (10), and let P be defined by (10). Then*

$$\mathbb{E}^{\mathbb{Q}}[P_t] = F(t) \quad \text{for all } t, \text{ and for every admissible} \\ (\kappa, \sigma_0^y, \sigma_1^y, m_0, m_1, \alpha_{01}, \alpha_{10}, \gamma_{01}, \gamma_{10}, \pi_0).$$

Proof. Taking expectations in (10) and using linearity, $\mathbb{E}^{\mathbb{Q}}[P_t] = F(t) + \mathbb{E}^{\mathbb{Q}}[X_t] - \mu_X(t) = F(t) + \mu_X(t) - \mu_X(t) = F(t)$. The content of the statement is not the algebra but the requirement that μ_X be the *same* functional of the parameters that drives the dynamics, evaluated by the same numerical procedure. Section 5.1 shows that $\mu_X(t) = u_0(t) + u_1(t)$, where u_i solve the coupled first-moment system (12). Since that system is integrated jointly with the regime probabilities used in pricing, the cancellation is exact in floating-point arithmetic up to integration tolerance, rather than approximate. $\square$

Remark 2. *Proposition 1 is what permits the identification argument of this paper. Whatever is wrong with the inherited regime parameters, and Section 7 shows that a great deal was wrong with one of them, none of it can corrupt the first moment. Errors in the residual layer are confined to the shape of the distribution around a forward level that remains pinned to observed quotes.*

Remark 3 (Contrast with a multiplicative specification). *An earlier specification of this project modelled the price directly as $P_t = s_P \sinh(Y_t)$ with Y a regime-switching OU process, for which $\mathbb{E}[P_t] = s_P e^{v(t)/2} \sinh(m(t))$ with v the transform-space variance. This form fails to anchor to the forward in two distinct ways depending on the mean-reversion rate. Under a near-unit-root κ the variance v grows almost linearly in horizon and the factor $e^{v/2}$ inflates the implied forward without bound. Under the corrected κ of Section 6 the inflation is mild, reaching only 1.019 at its stationary value, but the regime-weighted mean $m(t)$ is negative, so the implied expected spot converges to approximately $-3,108$ TRY/MWh against an observed forward of $+2,916$ TRY/MWh. Neither failure is a parameter problem that better estimation would fix; both follow from letting the residual dynamics determine the implied forward. The additive centred form removes the question by construction, which was the original motivation for adopting it. We retain the comparison in Section 7 because it quantifies what the centring buys.*

5. Valuation

5.1. The coupled moment system

Pricing requires $\mu_X(t)$ for the centring in (10) and the conditional distribution of X_T for the payoff. Both follow from a system of regime-partitioned moments. Define

$$p_i(t) = \mathbb{Q}(S_t = i), \quad u_i(t) = \mathbb{E}^{\mathbb{Q}}[X_t \mathbf{1}_{\{S_t=i\}}], \quad w_i(t) = \mathbb{E}^{\mathbb{Q}}[X_t^2 \mathbf{1}_{\{S_t=i\}}], \quad (11)$$

for $i \in \{0, 1\}$. Applying the Kolmogorov forward equation for the chain and Itô's lemma to (8) on each regime-restricted expectation yields a closed six-dimensional linear system. The property that matters for what follows is its recursive structure: p evolves autonomously, u depends on p , and w depends

on both, so the system is linear and no closure approximation is required.

$$\begin{aligned}
\dot{p}_0 &= -q_{01}p_0 + q_{10}p_1, \\
\dot{p}_1 &= q_{01}p_0 - q_{10}p_1, \\
\dot{u}_0 &= \kappa(m_0p_0 - u_0) + a_0p_0 - q_{01}u_0 + q_{10}u_1, \\
\dot{u}_1 &= \kappa(m_1p_1 - u_1) + a_1p_1 + q_{01}u_0 - q_{10}u_1, \\
\dot{w}_0 &= 2\kappa(m_0u_0 - w_0) + 2a_0u_0 + (\sigma_0^X)^2 p_0 - q_{01}w_0 + q_{10}w_1, \\
\dot{w}_1 &= 2\kappa(m_1u_1 - w_1) + 2a_1u_1 + (\sigma_1^X)^2 p_1 + q_{01}w_0 - q_{10}w_1,
\end{aligned} \tag{12}$$

initialised at $p_i(0) = \pi_i$, $u_i(0) = x_0\pi_i$, $w_i(0) = x_0^2\pi_i$, where π is the filtered regime distribution at the valuation hour and $x_0 = 0$. The terms in a_i carry the risk-neutral drift perturbation of Section 5.4 and are zero in the baseline.

The unconditional moments follow by summation,

$$\mu_X(t) = u_0(t) + u_1(t), \quad \text{Var}^{\mathbb{Q}}[X_t] = w_0(t) + w_1(t) - \mu_X(t)^2, \tag{13}$$

which establishes the representation used in the proof of Proposition 1.

One numerical point is worth stating here because it was a failure mode in development. The coefficients q_{01}, q_{10} and σ_i^X vary along the covariate path, so (12) is non-autonomous and cannot be advanced by matrix exponentials on a coarse grid; on hourly steps the second-moment trajectory was visibly unstable. Appendix Appendix C gives the integration scheme adopted in response.

5.2. Pricing equation

Conditional on the regime, the value function $V_i(x, t)$ for regime i solves the coupled system

$$\frac{\partial V_i}{\partial t} + [\kappa(m_i - x) + a_i] \frac{\partial V_i}{\partial x} + \frac{1}{2} (\sigma_i^X(t))^2 \frac{\partial^2 V_i}{\partial x^2} - rV_i + q_{ij}(t) (V_j - V_i) = 0, \quad j \neq i, \tag{14}$$

with terminal condition $V_i(x, T) = (F(T) + x - \mu_X(T) - K)^+$ for a call. The coupling term $q_{ij}(V_j - V_i)$ is the regime-switching contribution familiar from Buffington and Elliott (2002) and Jobert and Rogers (2006); the numerical treatment follows the finite-difference approach of Boyle and Draviam (2007) and Huang et al. (2011). We discretise x on a uniform grid of 1,201 nodes spanning a domain wide enough that the terminal density is numerically

negligible at the boundaries, and we report the option value as the π -weighted mixture $V_0(0, 0)\pi_0 + V_1(0, 0)\pi_1$.

The drift term in (14) and the a_i terms in (12) must be populated from the same parameter field. This is a genuine hazard rather than a housekeeping remark: adding a perturbation to the moment system alone, or to the pricing equation alone, breaks Proposition 1 silently, producing prices that remain plausible while no longer being anchored to the forward. We guard against it with a regression test that asserts $\max_t |\mathbb{E}^\mathbb{Q}[P_t] - F(t)| < 10^{-8}$ TRY/MWh across asymmetric, mixed-sign perturbation pairs, and which fails if the two code paths are ever allowed to diverge.

5.3. Independent verification

Two checks discipline the numerics. First, put–call parity, $C - P = e^{-rT}(F(T) - K)$, holds by no-arbitrage independently of the model and provides an internal consistency test. Across the full strike–maturity grid of Section 7.2 the maximum absolute parity error is 1.0×10^{-6} TRY/MWh, which is solver precision.

Second, (8) is simulated directly with an exact conditional Ornstein–Uhlenbeck step on the sampled regime path, 4×10^4 paths at a 0.05-hour time step, and compared against the finite-difference solution across strikes (Figure 3). The two routes share only the parameter file: one integrates a deterministic moment system and solves a boundary-value problem, the other samples trajectories. They agree to within roughly 1% near the money. The residual discrepancy is systematic rather than random, with Monte Carlo below the finite-difference value at every strike and the gap widening from 0.6% at $K = 2,000$ to 3.1% at $K = 4,000$. This is the signature of time discretisation in the sampled regime path: switching events occurring within a step are missed, which slightly understates mixing into the high-volatility state and therefore the deep out-of-the-money tail on which far strikes depend. Since the effect is concentrated where option values are smallest in absolute terms, it does not affect the conclusions drawn here, but it does mean the Monte Carlo figures should be read as a consistency check rather than as an independent valuation.

Agreement of this kind is informative about implementation correctness. It says nothing about whether the parameters describe the market.

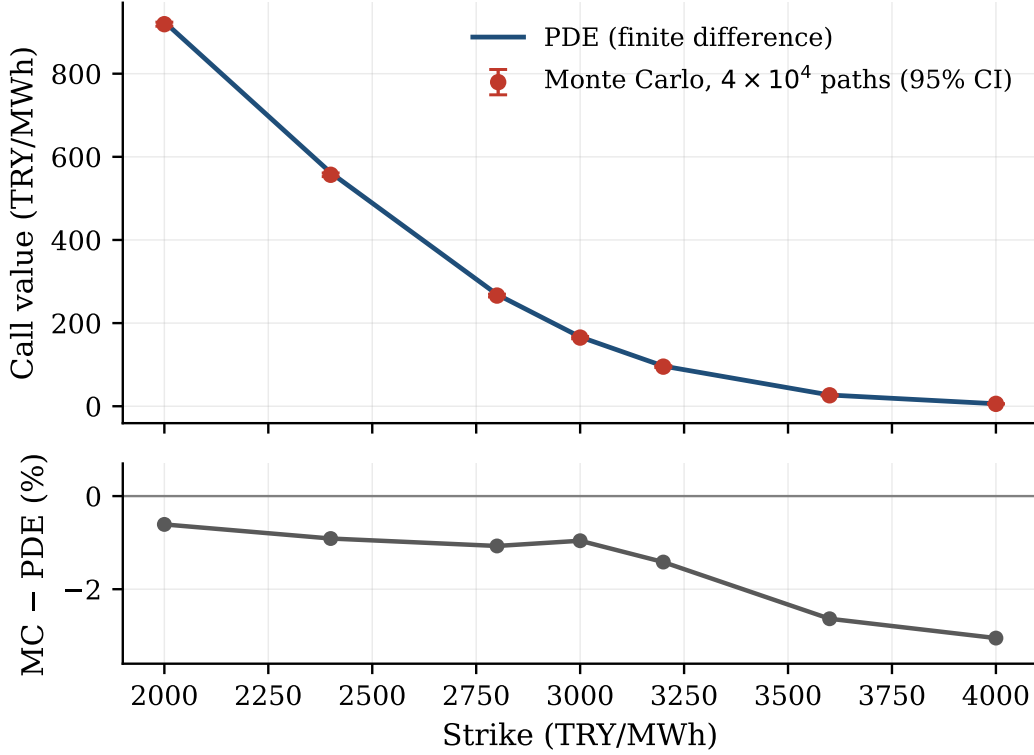


Figure 3: Finite-difference against Monte Carlo call values at 72 hours, with the relative difference below.

5.4. The risk-neutral drift channel

The regime and volatility parameters of Section 4 are historical estimates under $\mathbb{P}$. Using them unchanged for valuation asserts that the market prices neither regime nor volatility risk. Since the monthly strip cannot identify either premium, we do not attempt to estimate them. Instead we implement the change of measure explicitly so that the assumption becomes visible and its consequences measurable.

Following the structure in Janczura (2014), we admit a regime-dependent drift perturbation, replacing (8) by

$$dX_t = [\kappa(m_{S_t} - X_t) + a_{S_t}] dt + \sigma_{S_t}^X(t) dW_t^{\mathbb{Q}}, \quad (15)$$

with $a = (a_0, a_1)$ expressed in TRY/MWh per hour. Setting $a = (0, 0)$ recovers the physical-measure baseline exactly, bit for bit, so every result in this paper that does not explicitly vary a is a zero-premium result and is labelled as such.

The perturbation enters (12) through $a_i p_i$ in the first moment and $2a_i u_i$ in the second, so the effect on the variance, and hence on option value, is second order in a . Appendix Appendix A derives the order and reports the numerical check, a log-log slope of 2.0000 across $a_1 \in \{-0.5, \dots, -20\}$. At magnitudes plausible for a drift premium the resulting price change falls below the solver’s resolution. The implication is specific: within this specification a *drift*-type premium is nearly unidentifiable from option values even if one were observed, because its channel is quadratically suppressed. A premium shifting transition intensities rather than drift would not be suppressed in the same way, and that extension is left open.

6. Parameter provenance

A valuation model is only as interpretable as the audit trail behind its inputs. We therefore classify every parameter into one of four categories and report the classification alongside the value itself.

Calibrated determined by observed market prices through the constrained optimisation of Section 3.

Inherited taken from a historical estimation exercise under $\mathbb{P}$ and used for valuation without a measure adjustment.

Derived reconstructed from summary statistics of that exercise because the primary estimate was unavailable, with no standard error attached.

Assumed chosen by the analyst on economic or pragmatic grounds.

Table 2 applies the scheme. Only the forward curve is calibrated. Everything governing the distribution around it is inherited, derived or assumed, which is the concrete form the incompleteness of this market takes.

Three entries require comment because they carry more interpretive weight than their appearance in a table suggests.

The intercepts α_{01}, α_{10} were not available from the original estimation output, which reported slope coefficients and average marginal effects but not intercepts. We reconstructed them by solving, over the full covariate sample, the two scalar equations that match the implied average departure rates to the reported mean regime durations of 3.76 and 7.56 hours. The reconstruction reproduces the reported stationary stress share to within 1%

Table 2: Production parameters and provenance. Only $F(\cdot)$ is identified by market prices.

Parameter	Value	Status	Note
$F(\cdot)$	curve	Calibrated	Exact on six monthly averages
Jan. anchor	<code>spot_to_next</code>	Assumed	No quote exists
κ (h^{-1})	0.078394	Inherited	Half-life 8.84 h
σ_0^y	0.0035348	Inherited	Calm regime
σ_1^y	0.0924067	Inherited	Stress regime
s_P (TRY/MWh)	282.48	Assumed	Source file unavailable
α_{01}	-1.015666	Derived	Duration reconstruction
α_{10}	-1.895229	Derived	Duration reconstruction
γ_{01}	-0.583778	Inherited	Ramp term omitted
γ_{10}	+0.077698	Inherited	Ramp term omitted
$\pi_{1,0}$	0.068023	Inherited	Filtered at valuation
a_0, a_1	0, 0	Assumed	Zero drift premium
r (annual)	0.40	Assumed	Flat, not calibrated

under an ergodic approximation and 4.5% under simulation along the realised covariate path. It is nevertheless a moment-matching reconstruction and not a likelihood estimate, so no standard error attaches to it and no inference should be drawn from its magnitude.

The slopes γ_{01}, γ_{10} come from a specification that also contained a residual-demand ramp covariate, which the production implementation does not carry. If the original fit was joint, our slopes are subject to omitted variable bias of unknown sign. We were unable to determine from the available artefacts whether the two covariates were estimated jointly or separately, and we report the ambiguity rather than resolving it by assumption.

The initial regime distribution π comes from a filter run under an earlier model specification than the one supplying the volatilities, so the production configuration mixes sources at the initialisation step. We quantify the consequence directly in Section 7.3 by re-pricing under the alternative of initialising at the stationary distribution.

7. Results

7.1. Curve reproduction and shape

The constrained solve reproduces all six monthly averages to a maximum absolute error of 3.6×10^{-12} TRY/MWh (Figure 4), which establishes only that the constraint set is satisfied, for the reason given in Section 3.

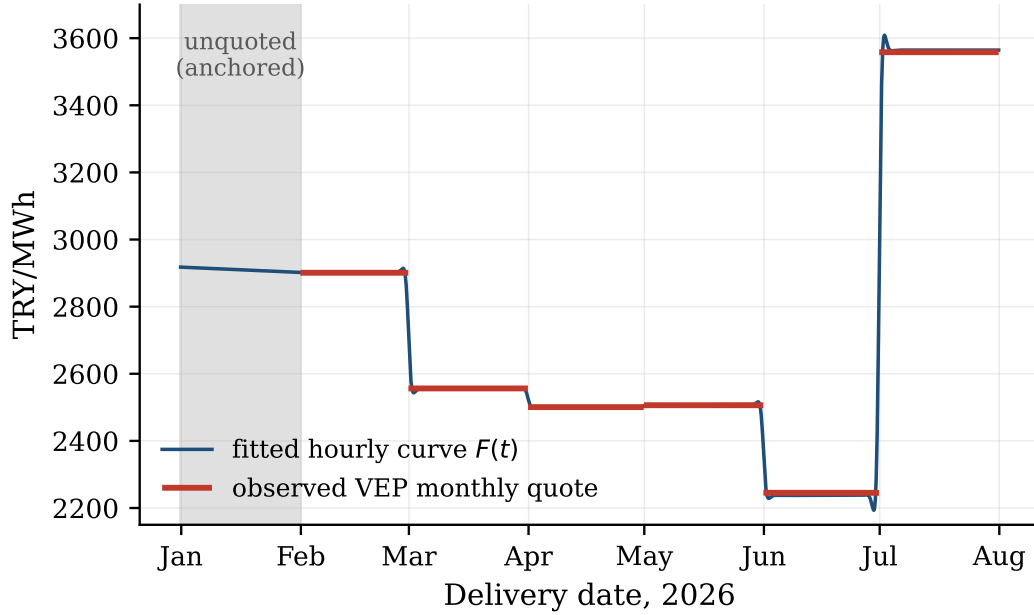


Figure 4: Fitted hourly forward curve against the observed monthly baseload quotes. Shading marks the unquoted January window.

The shape of the curve is more informative than the fit. Hourly increments have a mean absolute size of 0.45 TRY/MWh, and the three largest increments over the whole horizon sit at month boundaries where the quote strip itself steps, the largest being +47.0 TRY/MWh at June–July. That step reflects the 58% rise between the June and July quotes and is therefore a property of the data rather than an artefact of interpolation. Within the January window the largest increment is 0.05 TRY/MWh, so the anchor joins the quoted region without a discontinuity, and no oscillation of the kind unconstrained spline interpolation can produce near sharp level changes is present.

7.2. Baseline valuation and the option surface

At the reference contract, a European call struck at 3,000 TRY/MWh expiring 72 hours after valuation, the model gives $F(T) = 2,916.16$ TRY/MWh, terminal residual standard deviation 532.26 TRY/MWh, and a value of 166.75 TRY/MWh by finite differences against 169.18 TRY/MWh by simulation. The simulated probability of a negative terminal price is zero at four decimal places, so the additive specification, while unbounded below in principle, is not operating near that boundary at the calibrated dispersion.

Table 3 reports the surface at three representative strikes, and Figure 5 displays the grid as strike and maturity sections. Sections are preferred to a perspective surface because exact values remain readable.

Table 3: Expiry-hour European option values from the production configuration. All expiries fall inside the unquoted January window. All quantities in TRY/MWh.

Maturity (h)	Strike	$F(T)$	Call	Put	Residual SD
24	2,000	2,917.24	927.11	10.87	524.08
	3,000	2,917.24	163.93	246.60	
	4,000	2,917.24	5.59	1,087.16	
72	2,000	2,916.16	924.70	11.55	532.26
	3,000	2,916.16	166.75	250.32	
	4,000	2,916.16	5.94	1,086.22	
336	2,000	2,910.21	907.92	11.57	531.19
	3,000	2,910.21	161.84	250.27	
	4,000	2,910.21	5.66	1,078.86	
720	2,000	2,901.51	883.95	11.60	529.61
	3,000	2,901.51	154.91	250.21	
	4,000	2,901.51	5.28	1,068.24	

One feature of the surface separates it from the term structure of a diffusive model and follows directly from the mean-reverting residual: terminal dispersion saturates rather than growing (Figure 6). With a half-life of 8.84 hours the residual reaches its stationary distribution within roughly a day, after which option value varies with maturity almost entirely through the forward level and the discount factor rather than through accumulated uncertainty. The mild decline in at-the-money value across the maturities in Table 3 is therefore attributable to the downward slope of the anchored Jan-

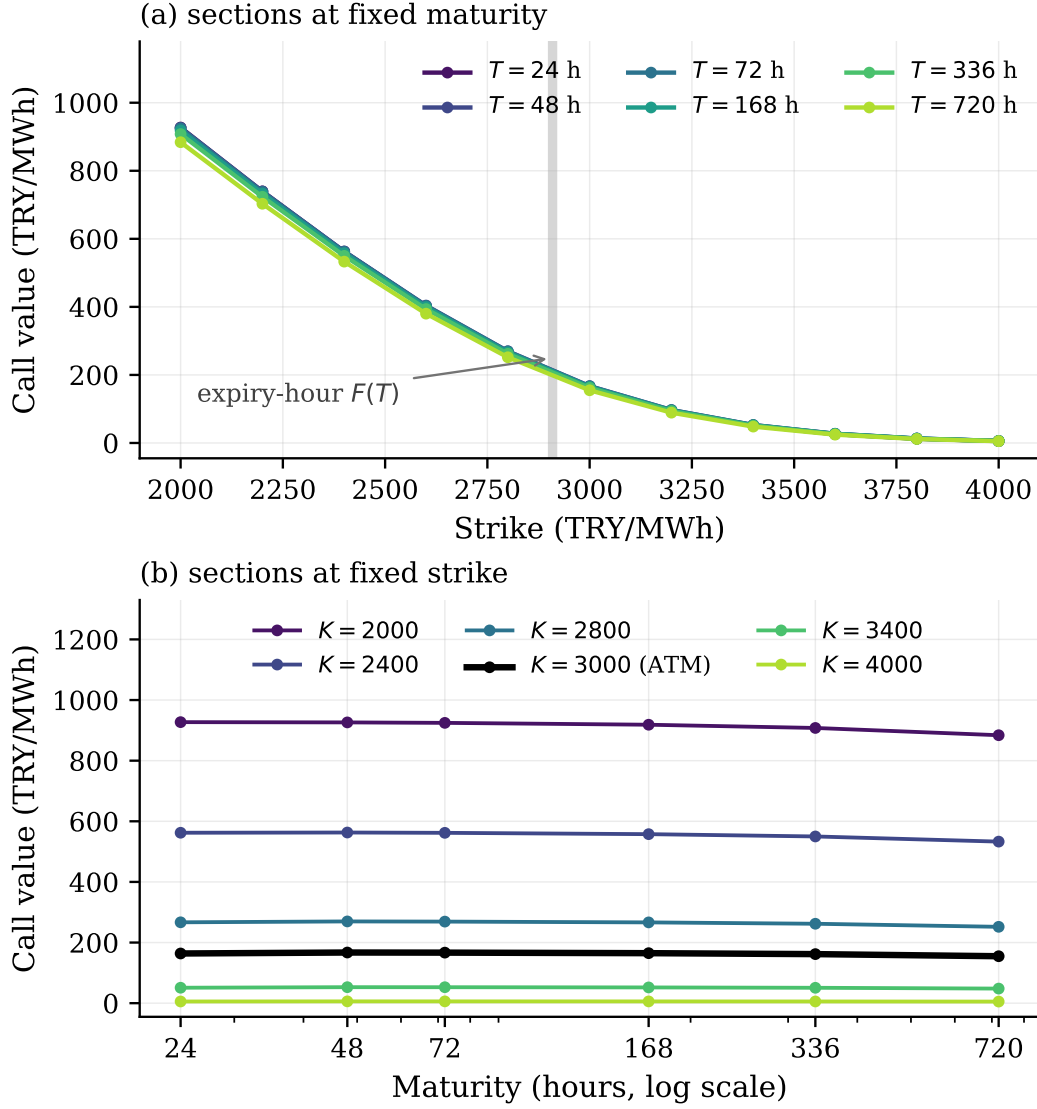


Figure 5: Two sections through the option price surface. Panel (a) shows call value against strike at each maturity, with shading marking the range of expiry-hour forward levels; panel (b) shows call value against maturity at representative strikes, the near-at-the-money strike emphasised.

uary curve rather than to any reduction in volatility.

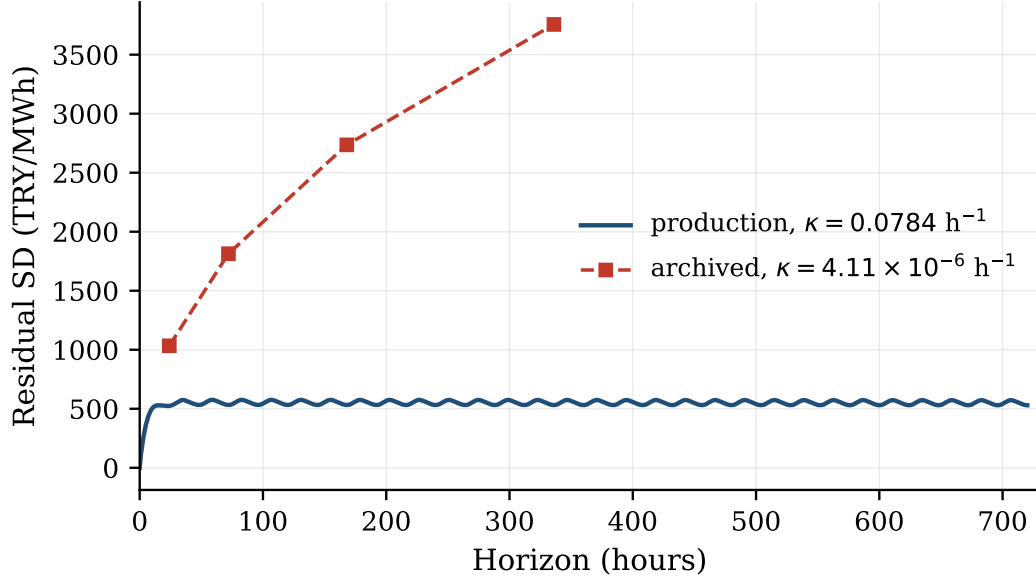


Figure 6: Terminal residual standard deviation against horizon, production against archived specification.

7.3. Sensitivity to unidentified inputs

Table 4 collects controlled experiments in which a single input is varied and everything else is held at production values. We order them by the magnitude of the resulting price change, because that ordering is itself the main result of this section.

The anchor dominates everything else.. Shifting the January anchor level by $\pm 20\%$, holding the curve shape rule fixed, moves the call by -89% to $+217\%$ (Figure 7); no other input comes within an order of magnitude. The mechanism is simple. The anchor sets the forward level at expiry, moneyness dominates the value of a near-at-the-money option, and the residual dispersion of roughly 530 TRY/MWh is small against a ± 580 TRY/MWh shift in the forward. The practical implication is direct: where the nearest delivery month is unquoted, the precision of any reported option value is bounded above by the uncertainty in the near-term forward level, and refinements to the residual specification cannot improve on that bound. A single reported price is then less informative than the anchor-conditional range.

Table 4: Controlled sensitivity of the 72-hour, $K = 3,000$ call. Each row varies one input; all others are held at production values. Baseline 166.75 TRY/MWh.

Input varied	Perturbation	Call (TRY/MWh)	Change
January anchor level	-20%	17.80	-89.3%
January anchor level	-10%	65.75	-60.6%
January anchor level	+10%	326.05	+95.5%
January anchor level	+20%	528.96	+217.2%
Mean-reversion rate	archived near-unit-root value	687.04	+312.0%
Residual demand path	-2σ offset	200.81	+20.4%
Residual demand path	$+2\sigma$ offset	107.97	-35.2%
Regime volatilities	alternative model estimate	165.58	-0.70%
Discount rate	$r = 0.15$	167.09	+0.21%
Discount rate	$r = 0.60$	166.47	-0.16%
Initial regime distribution	stationary rather than filtered	166.75	+0.0003%
Drift premium a_1	plausible magnitudes	166.75	$< 10^{-4}\%$

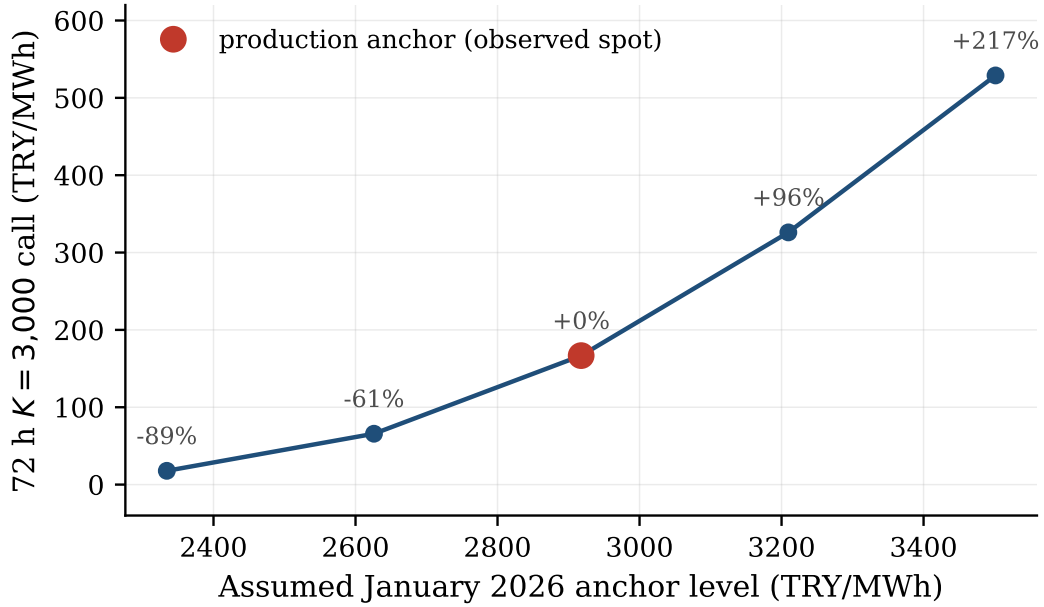


Figure 7: Option value against the assumed January anchor level, curve shape rule held fixed.

Persistence is the dominant residual-layer input.. The archived specification inherited its mean-reversion rate from an autoregressive fit to the untransformed price level, giving $\phi = 0.999996$, $\kappa = 4.11 \times 10^{-6} \text{ h}^{-1}$ and a within-regime half-life near nineteen years. Under that value the same contract prices at 687.04 TRY/MWh, more than four times the production figure, because a near-unit-root residual accumulates variance almost linearly in horizon instead of saturating. The correction substitutes the deseasonalised single-regime coefficient $\phi = 0.9246$, giving $\kappa = 0.0784 \text{ h}^{-1}$ and the saturating dispersion path of Figure 6.

The two figures are not in conflict once the objects are separated. A Markov-switching AR(1) has a marginal autocorrelation that mixes a fast component, decaying at rate $q_{01} + q_{10} \approx 0.398 \text{ h}^{-1}$, with a slow component decaying at $-\ln \phi$. Because the regime-conditional means here are small and close together, the regime-shift contribution to variance is around 5%, so the marginal autocorrelation is dominated by the slow component. A 200,000-hour simulation confirms an empirical half-life consistent with the near-unit-root value to within 2% of theory. The nineteen-year figure correctly describes persistence of the raw transformed price, which over the estimation window absorbs lira depreciation; it does not describe persistence of deviations around a forward curve, which is the object (8) requires. Diagnosing this took separating two quantities that a single reported statistic had conflated.

What the centring buys.. Running the same parameter set through the archived multiplicative specification of Remark 3, which lets the residual dynamics determine the implied forward rather than anchoring it, gives Figure 8. The centred form returns $F(T)$ at every horizon. The multiplicative form, on identical parameters, converges instead to an implied expected spot near $-3,108$ TRY/MWh against an observed forward of $+2,916$ TRY/MWh. The sign arises from the regime-weighted mean in transform space rather than from the variance inflation dominating under the archived mean-reversion rate, so the specification fails differently at different parameter values while failing in both. This is the practical content of Proposition 1: forward consistency is either imposed or it is not obtained.

The covariate channel is economically signed.. Shifting the residual demand path by -2σ raises the call by 20.4% and by $+2\sigma$ lowers it by 35.2% (Figure 9), with terminal dispersion moving from 608 to 404 TRY/MWh across

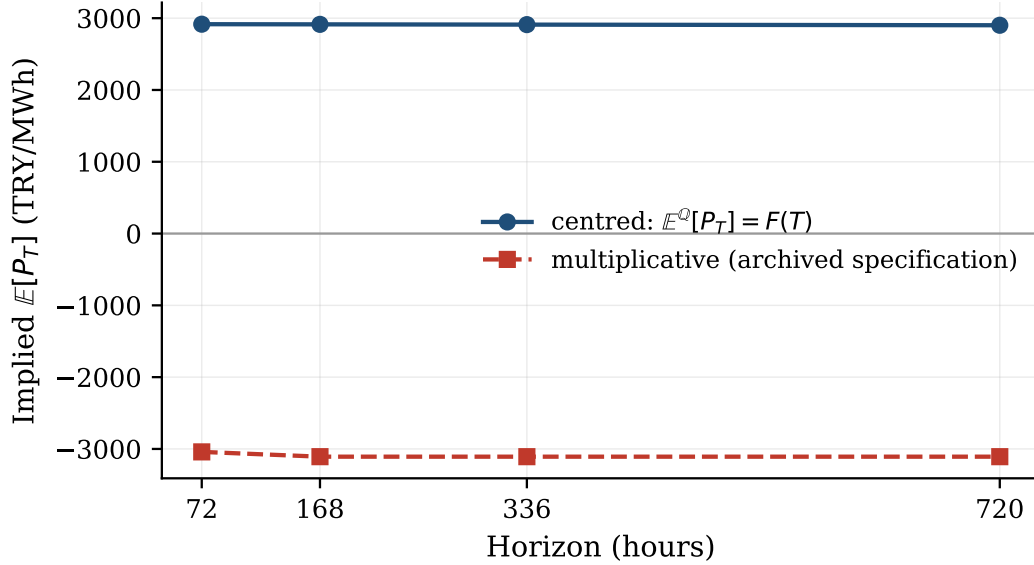


Figure 8: Implied expected spot, centred against archived multiplicative specification.

the same range. The sign follows from the estimated slopes: $\gamma_{01} < 0$ means high residual demand suppresses transitions into the high-variance regime in this specification.

That direction runs against the intuition that scarcity drives volatility, and it deserves comment. Two considerations make it less surprising than it first appears. The regimes here are variance states in transform space rather than economic stress states, as Appendix Appendix B notes, so the relevant question is when hourly price variability is large rather than when prices are high. And low residual demand is precisely the condition under which renewable output displaces the flexible thermal units that would otherwise absorb short-run imbalances, leaving a supply stack with less capacity to respond smoothly. The realised sample is consistent with this reading: across the seven evaluation months, the correlation between the monthly mean price and the monthly standard deviation of realised prices is -0.67 , with the highest-price month showing the lowest dispersion (573 TRY/MWh) and one of the lowest-price months the highest (1,258 TRY/MWh). With seven observations this is suggestive rather than conclusive, and we do not treat it as a test.

The direction is therefore reported as a property of the estimated co-

efficients, not as an established economic finding, since the omitted ramp covariate documented in Section 6 could alter it and the intercepts entering the same logistic functions are reconstructions rather than estimates.

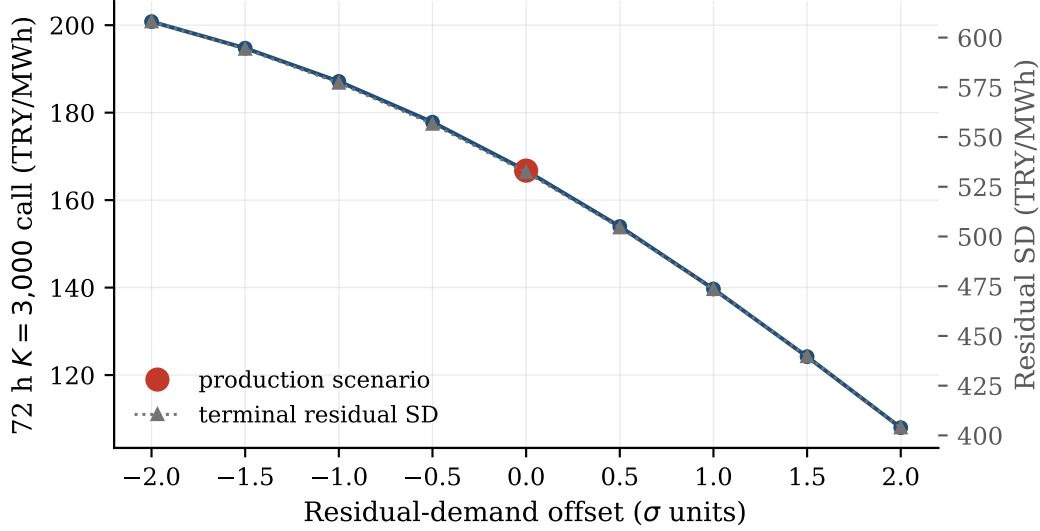


Figure 9: Call value and terminal dispersion across a $\pm 2\sigma$ range of residual-demand offsets.

Everything else is second order.. Substituting the regime volatilities of the closest competing estimated model changes the price by 0.70%, despite a large difference in fit criteria between the two specifications. The discount rate is similarly immaterial across $r \in [0.15, 0.60]$, a range bracketing Turkish policy rates over the period: at 720 hours the factor e^{-rT} moves only from 0.988 to 0.952. That is a property of the maturities studied rather than evidence that the assumption is innocuous in general, and contracts extending over several months would require the term structure of rates to be modelled rather than assumed. Initialising the chain at its stationary distribution rather than the filtered one is likewise immaterial at 72 hours, since with mean regime durations under eight hours the chain has forgotten its initial condition well before expiry; the same substitution moves a 6-hour option by roughly 29%, so the irrelevance again belongs to the horizon rather than to the parameter. The drift premium channel is quadratically suppressed, as established in Section 5.4.

7.4. Realised evaluation

The forward curve constructed at the end of 2025 can be compared against what subsequently occurred. Table 5 reports, for each delivery month, the realised mean, the bias of the curve against it, the mean absolute error, the symmetric percentage error, and a dispersion ratio comparing mean model residual standard deviation to the realised standard deviation of realised-minus-forward within that month. Figure 10 plots the two series as daily means together with the monthly biases, the January bar distinguished because its level comes from the anchor rather than from a quote.

Table 5: Realised evaluation against the production forward curve, January to July 2026.

Month	Hours	Realised mean	Bias	MAE	sMAPE	Disp. ratio
2026-01 [†]	744	2,894.92	−14.47	451.93	16.8%	0.97
2026-02	672	2,078.20	−822.79	992.40	50.0%	0.56
2026-03	744	1,620.32	−935.99	1,193.58	71.5%	0.46
2026-04	720	921.06	−1,579.60	1,873.41	126.3%	0.41
2026-05	744	590.90	−1,915.35	2,150.69	155.2%	0.44
2026-06	720	1,240.16	−1,005.17	1,458.19	101.3%	0.35
2026-07	744	2,699.61	−858.58	1,058.33	40.9%	0.61
Overall	5,088	1,721.72	−1,019	1,312.39	80.4%	—

[†] January carries no observed quote; its curve level is the near-term anchor assumption.

Three observations follow, and we state them in order of how much they constrain the interpretation of the rest of the paper.

First, the forward strip missed badly. Realised prices fell below the quoted forward in every month from February onwards, by 24% in July and by 76% in May, with an overall mean bias of −1,019 TRY/MWh. In May, 222 of 744 delivery hours cleared below 10 TRY/MWh. The magnitude of the deviation is consistent with a large supply-side shock rather than with ordinary forecast error, and the period coincided with exceptionally favourable hydrological conditions and record renewable output that displaced gas-fired generation from the merit order. We describe the coincidence and do not estimate it; attributing the deviation causally would require a supply-stack decomposition that we do not perform, and the merit-order mechanism is well documented elsewhere (Sensfuß et al., 2008; Ketterer, 2014; Clò et al., 2015; Sirin and Yilmaz, 2020).

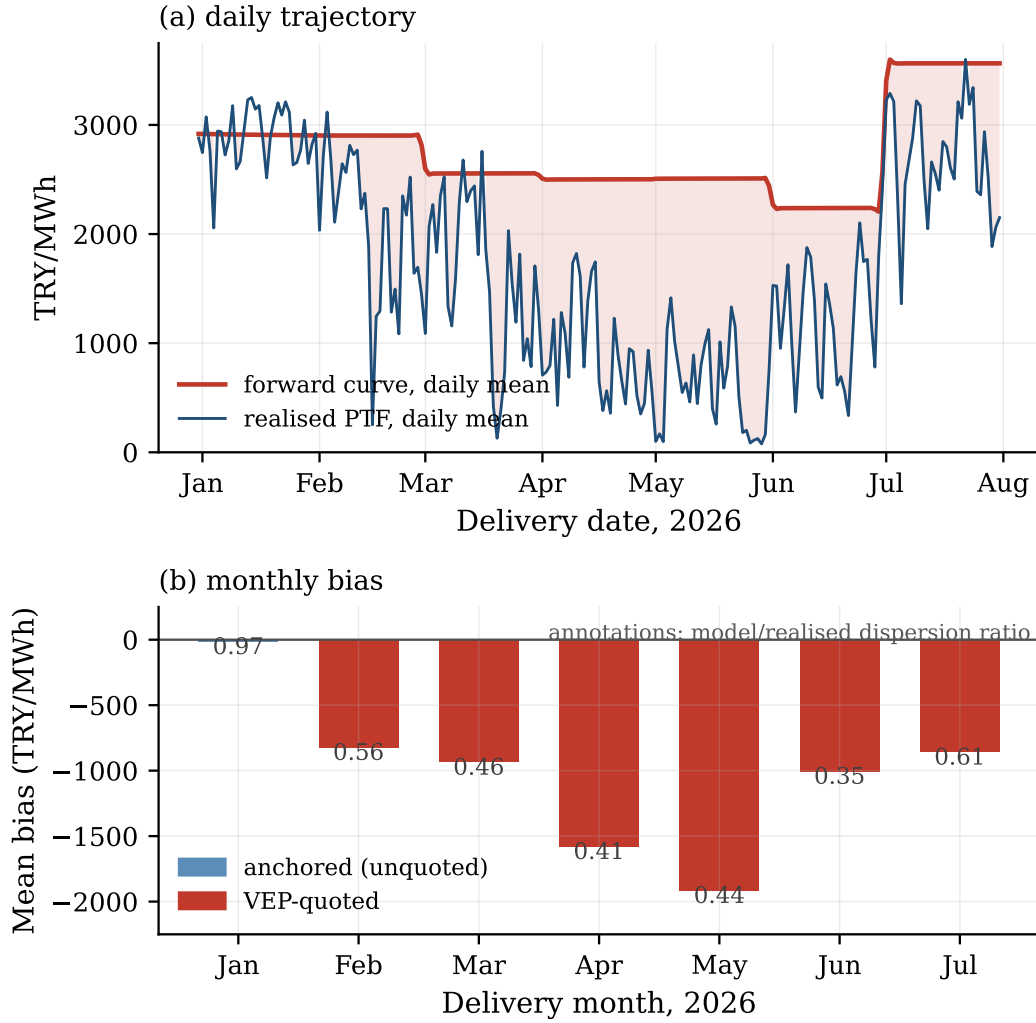


Figure 10: Realised evaluation. Panel (a) plots realised PTF against the production forward curve as daily means; panel (b) gives the monthly mean bias, with the anchored January window distinguished from the VEP-quoted months.

Second, because our curve reproduces the quotes exactly, the entire level error is inherited from the forward market. No property of the residual specification, the regime structure or the pricing scheme contributed to it. The error therefore bounds what forward-anchored valuation can deliver in this market, and the bound is not a small one: an option value anchored to a forward curve that is subsequently wrong by 40% on average is wrong by a correspondingly large amount, however carefully the distribution around the curve was specified. This is a structural property of the anchoring approach rather than a defect of our implementation, and it applies equally to any of the curve-construction methods cited in Section 3.

Third, the month that was not quoted performed best. January, whose level came entirely from the near-term anchor assumption, shows a bias of -14.47 TRY/MWh and the lowest sMAPE in the sample, while every quoted month missed by hundreds of lira. We caution against reading this as evidence that the anchor rule outperforms the market. The anchor was set at the observed spot and interpolated towards February, so it inherited the information content of a price observed one hour before the window opened, whereas the quoted months encode expectations formed up to seven months ahead. The comparison illustrates how rapidly forward information decayed during this episode rather than the merits of an interpolation rule.

7.5. Evidence from seven valuation dates

A single valuation date cannot distinguish an unusual episode from a persistent property of the quote strip. The curve construction can be replayed at earlier dates without contamination, because it uses only the quotations published on the valuation date and the spot of that hour and contains no historical parameter. The residual dynamics cannot be replayed, since their parameters are fitted to data through 2025, so this exercise evaluates the forward curve alone and prices no option. Seven valuation dates at six-month intervals from 2022-12-31 to 2025-12-31 yield 49 (date $\times$ delivery month) observations. Replaying 2025-12-31 through the same code reproduces the published curve to 4.5×10^{-13} TRY/MWh over 5,089 hours.

Three results follow, and the first revises the reading of Section 7.4.

The front month is never quoted.. The anchor rule fires at all seven dates. Every strip prices delivery months two through seven ahead and none prices the first, so the structural claim of Section 3.3 holds across three years rather than resting on one observation.

2026 is large but not exceptional.. The $-1,019$ TRY/MWh mean bias of the 2025-12-31 curve sits at the 32.7th percentile of the pooled distribution and the 28.6th percentile of the seven date means, inside their interquartile range. Two earlier dates produced more negative outcomes, the worst being 2022-12-31 at $-1,465$ TRY/MWh (-39.8%). The hydrological episode of 2026 is therefore an instance of a magnitude the strip has produced before, not a departure from otherwise reliable behaviour.

The bias is large, negative on average, and unstable in sign.. Pooled across dates the mean bias is -603 TRY/MWh (-18.2% in relative terms) with a standard deviation of 775 TRY/MWh. Realised prices came in below the strip at five dates and above it at two, with date means running from $-1,465$ to $+290$ TRY/MWh (Figure 11a). The direction is not a stable property. Statistical support for a non-zero mean is correspondingly weak once dependence is respected: pooling all 49 observations gives $t = -5.45$, but the same delivery month appears from several valuation dates and consecutive months of one curve share a market state, so that figure is anti-conservative. Aggregating to one observation per date gives $t = -2.29$ on seven dates, $p = 0.062$, and restricting to non-overlapping delivery windows leaves p between 0.11 and 0.49. We therefore report the mean bias as economically large and describe the strip as a biased and unstable predictor, without claiming a calibrated significance level on seven dates.

Table 6: Mean bias of the forward curve against realised monthly baseload, by valuation date. Seven delivery months per date.

Valuation date	Mean bias (TRY/MWh)	Median (TRY/MWh)	Relative (%)
2022-12-31	$-1,464.63$	$-1,545.59$	-39.8
2023-06-30	$-1,250.01$	$-1,299.88$	-36.5
2023-12-31	-756.89	-813.27	-24.5
2024-06-30	-174.47	-214.48	-6.2
2024-12-31	$+151.28$	$+179.60$	$+7.2$
2025-06-30	$+289.76$	$+307.18$	$+11.4$
2025-12-31	$-1,018.85$	-935.99	-39.1
Pooled ($n = 49$)	-603.40	-353.70	-18.2

The anchor band has approximately the right scale.. The $\pm 20\%$ band used in Section 7.3 was chosen without empirical support. Across the seven anchored

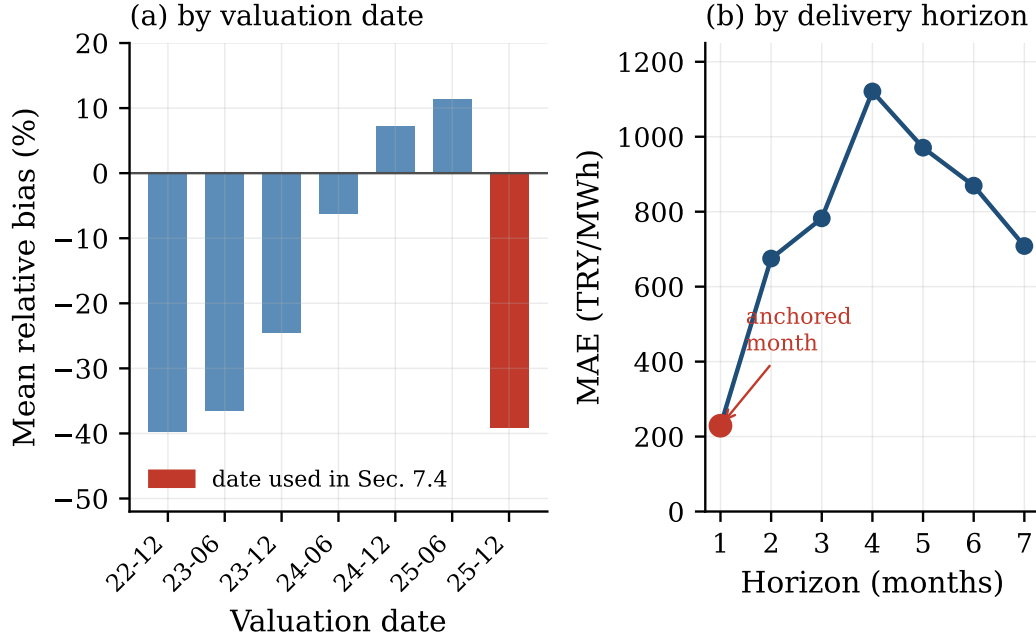


Figure 11: Forward curve error across seven valuation dates. Panel (a) gives the mean relative bias per date; panel (b) the mean absolute error by delivery horizon.

months the realised relative error has a standard deviation of 11.7%, against 11.5% for a uniform distribution on $\pm 20\%$, so the band is of the right width. It is not a containment bound: one of seven anchored months falls outside it, at -21.2% . And its centre is displaced, the empirical mean being -3.5% , so the linear ramp over-prices the anchored month slightly on average. The band should be read as a scale for anchor uncertainty rather than as a range that brackets it.

Error does not grow monotonically with horizon.. Figure 11b shows mean absolute error rising from 229 TRY/MWh at the anchored first month to 1,121 TRY/MWh four months ahead, then falling back to 708 TRY/MWh at seven, with a Spearman rank correlation of +0.25 between horizon and absolute error. The decline at the long end should not be read as skill. The first horizon is always the anchored month, so its small error reflects a ramp from an observed spot rather than a quoted forward, and with dates six months apart the seventh horizon is always the same pair of calendar months, making the long end as much seasonal as maturity-driven.

Four of the seven valuation dates fall on a non-business day, when no

index price is published; for those the strip is the last publication before the date, one to four days stale, while the spot remains that of the valuation night. No later publication is used anywhere, the affected observations are flagged in the released data rather than dropped, and a strict same-day rule would reduce the study to three dates.

A fourth observation concerns the model’s own dispersion. The ratio of model residual standard deviation to realised dispersion is 0.97 in January and falls to 0.35–0.61 thereafter, so outside the anchored window the model is too narrow by a factor of roughly two. Part of this is expected: the production residual is a single-factor approximation, and a companion two-factor specification developed alongside this work carries an additional slow component that the production configuration does not represent. Part is the shock itself, which widened realised dispersion beyond anything the historical sample contains. We do not attempt to separate the two here.

8. Discussion

8.1. *What a monthly strip identifies*

The results support a narrow reading of what calibration to a baseload strip accomplishes. It fixes six linear functionals of a curve with several thousand degrees of freedom, and through Proposition 1 it fixes the first moment of the price distribution at every horizon. It does not fix the intraday shape of the curve, the level over unquoted months, the volatility of deviations around the curve, or any risk premium. The sensitivity ordering in Table 4 makes the consequence concrete: the largest single driver of the reported option value is an input that no observed price constrains.

A reporting convention follows. Rather than a point value, valuation in markets of this type is better expressed as a value conditional on an explicitly stated anchor, accompanied by the range induced by plausible alternatives. The $\pm 20\%$ anchor band used here produces a call range of 17.80 to 528.96 TRY/MWh around a central 166.75 TRY/MWh, and that interval carries more information than its midpoint. Section 7.5 supplies the band with a warrant it previously lacked: across seven valuation dates the realised anchor error has a standard deviation of 11.7%, close to the 11.5% a uniform $\pm 20\%$ prior implies, though displaced by -3.5% from zero and not containing the tail.

8.2. What the strip predicts, and what that costs

The multi-date evidence bears on a question wider than valuation mechanics. Whether forward prices are unbiased predictors of subsequent spot outcomes is an old subject in electricity markets, where limited storage and inelastic demand make the hedging-pressure equilibrium of Bessembinder and Lemmon (2002) a natural frame and where realised premia have been measured for mature markets by Longstaff and Wang (2004) and Algieri et al. (2021). On Turkish monthly baseload contracts over 2023–2026, the answer is that the strip is neither unbiased nor reliably biased in one direction. Its pooled error averages -18% of the forward level, so a buyer of the strip paid on average well above what delivery subsequently cost; but two of seven dates reversed the sign, and the date-level mean is not distinguishable from zero at conventional levels once the dependence between overlapping windows is respected.

For anyone using such a curve, the practical consequence is about capital rather than about point forecasts. A dispersion of 775 TRY/MWh around a mean of -603 TRY/MWh on a price level near 2,500 TRY/MWh means the position risk of a forward-anchored book is dominated by the level of the curve it is anchored to, not by the shape of the distribution placed around that level. Provisioning against the residual volatility of Section 7, terminal dispersion of roughly 530 TRY/MWh at the horizons studied, while treating the anchor level as known would understate the exposure by a wide margin. The ordering in Table 4 says the same thing from the pricing side, and the multi-date distribution is what puts a number on it.

We stop short of decomposing the error into a risk premium and a forecast error. Separating the two requires either an instrument for expectations or a structural model of the supply stack, and seven valuation dates would not identify it. That the strip was systematically above realised outcomes in the years when hydrology was favourable, and below in the years when it was not, is consistent with both a premium and a persistent forecasting difficulty, and this sample cannot distinguish them.

8.3. Separation of concerns as a design property

The centring identity does more than tidy the algebra. It creates a firewall between the market-identified and the assumed parts of the model, and the value of that firewall emerged in a way that had not been anticipated when it was adopted. When the inherited mean-reversion rate was found to be wrong by four orders of magnitude, the correction changed option values

by a factor of four while leaving the forward reproduction, the monthly fit errors and the realised level bias unchanged. In a specification where residual dynamics fed back into the implied forward, everything would have required recalibration and the diagnosis would have been considerably harder, since the symptom would have surfaced in the calibration layer rather than only in the dispersion.

The same property holds for the measure change. Because the drift perturbation enters both the moment system and the pricing equation through a shared field, $\mathbb{E}^Q[P_t] = F(t)$ survives any (a_0, a_1) , verified numerically at magnitudes far outside any economically plausible range. Risk premium assumptions can therefore be explored without disturbing the calibration, which is what makes them reportable as sensitivities rather than as a separate model.

8.4. Persistence, transformation and nominal prices

The mean-reversion episode carries a methodological lesson that generalises beyond this dataset. Reported persistence statistics from a regime-switching estimation describe several distinct objects: within-regime autoregressive persistence, the marginal persistence of the observed series, and the persistence of deviations around a reference curve. These coincide only under restrictive conditions. Where the estimation sample also spans substantial currency depreciation, the raw series carries a monetary trend that inflates measured persistence further, and a coefficient transferred from that context into a residual-dynamics role will overstate memory by whatever the trend contributes. Diagnosis here required reconstructing the marginal autocorrelation of the switching process analytically and verifying it by simulation, since neither reported statistic alone identified the problem.

This is likely a common failure mode wherever parameters are handed between modelling stages, and particularly so in emerging markets where nominal series are non-stationary for reasons unrelated to the physics of the power system. The defence that eventually worked is worth stating as a rule: every transferred parameter should name the process it was estimated on, and any mismatch between that process and its intended use should be treated as an error until shown otherwise.

9. Limitations and future work

Identification. Volatility and regime risk premia are not identified by a monthly baseload strip, and no attempt is made here to estimate them.

All reported values are zero-premium values in the sense of Section 5.4. A drift-type premium is additionally quadratically suppressed within this specification, so even option data might identify it only weakly; an intensity-type premium is the more promising target and is not implemented.

Parameter provenance.. Three items in Table 2 carry unresolved questions. The transition intercepts are moment-matching reconstructions without standard errors. The transition slopes come from a specification containing a ramp covariate that the production implementation omits, with omitted-variable consequences of unknown sign. The transform scale $s_P = 282.48$ could not be verified because the preprocessing metadata defining it was unavailable; it is a median absolute price over a training window spanning years of substantial lira depreciation, so the price regime it represents may differ materially from that at the valuation date. Each would be resolved by access to the original estimation artefacts.

Specification scope.. The production residual is a single-factor mean-reverting process, unbounded below, whereas realised prices face administrative floors and ceilings. A companion specification developed alongside this work adds a slow second factor and explicit price limits. It is not integrated here because clipping the state distribution breaks the centring identity of Proposition 1: restoring $\mathbb{E}^\mathbb{Q}[P_t] = F(t)$ under a clipped law requires redesigning the centring term rather than adding a bound to the existing one. The trade-off is worth stating explicitly, because the obvious alternative is worse. Multiplicative or log-type specifications respect the lower bound automatically, but under Jensen’s inequality their first moment depends on the residual variance, so forward consistency can no longer be imposed algebraically. Figure 8 quantifies what that costs on this data, and the forward inconsistency it shows is larger by orders of magnitude, measured in TRY/MWh, than the boundary violations the additive form permits, whose simulated probability at the calibrated dispersion is zero to four decimal places. That is the most substantive extension available and is treated as the natural next study rather than a patch to this one. The dispersion ratios in Table 5 indicate what a second factor would need to supply. The price limit schedule used by the companion specification was inferred from realised price behaviour rather than read from a primary regulatory document, and is marked accordingly; it plays no role in the production results.

Evaluation.. The option-level evaluation still rests on a single valuation date, because the residual parameters cannot be replayed at earlier dates without look-ahead bias. Only the forward curve is evaluated across the seven dates of Section 7.5, and the two exercises therefore answer different questions. Seven dates six months apart also leave the delivery windows partially overlapping and the long horizons confounded with calendar month, so the horizon profile in particular should be read as descriptive. Four of the seven strips were published one to four days before their valuation date because the date fell on a non-business day; a strict same-day rule would reduce the sample to three. Extending the panel as further quotations and outturns accumulate, and re-estimating the residual process at each date so that option prices could also be evaluated out of sample, are the natural continuations.

10. Conclusion

A valuation framework has been developed for expiry-hour electricity options in a market that supplies a sparse monthly forward strip and no option prices. Its organising principle is a separation between what such a market identifies and what it does not. An equality-constrained hourly curve carries the identified information, a regime-switching residual with residual-demand-driven transition probabilities carries the assumed distributional structure, and a centring term ensures the two meet without the second contaminating the first: $\mathbb{E}^{\mathbb{Q}}[P_t] = F(t)$ holds for every admissible parameter vector, as an algebraic identity rather than a calibration target.

Applying the framework to Turkish data yields three findings. The dominant source of uncertainty in the reported option value is the forward level over the unquoted nearest delivery month, which overwhelms every parameter of the residual specification by an order of magnitude. The mean-reversion rate is the dominant residual-layer input, and transferring it from an autoregression on the untransformed price level rather than on deviations around the curve inflated option values by a factor of four until diagnosed. And the forward strip itself is a poor predictor of what follows: across seven valuation dates the mean error is -18% of the forward level, negative at five dates and positive at two, with the much-discussed 2026 outcome lying inside the interquartile range rather than outside it.

The last of these deserves emphasis. Forward-anchored valuation inherits the predictive accuracy of the forward market entirely, and on this evidence that accuracy is low and unstable in sign for reasons no pricing model can

correct. Combined with the finding that the nearest delivery month is never quoted, the implication for practice is that a single option value in this market conveys less than the value reported together with the range implied by the inputs no price identifies. What that discipline looks like in practice is what this paper has tried to demonstrate, including where it revealed our own errors.

Appendix A. Derivation of the moment system

Let S be the two-state chain with generator $Q(t)$ from (7) and let X solve (15). Write $f_i(x, t)$ for the joint density of $(X_t, S_t = i)$. The forward equation for the pair is

$$\partial_t f_i = -\partial_x[(\kappa(m_i - x) + a_i) f_i] + \frac{1}{2} (\sigma_i^X)^2 \partial_{xx} f_i + q_{ji} f_j - q_{ij} f_i, \quad j \neq i. \quad (\text{A.1})$$

Multiplying by 1, x and x^2 and integrating over $x \in \mathbb{R}$, with the boundary terms vanishing because f_i and xf_i decay at infinity, gives the system (12) in the variables $p_i = \int f_i dx$, $u_i = \int xf_i dx$ and $w_i = \int x^2 f_i dx$. The three blocks are recursively coupled: p is autonomous, u depends on p , and w depends on u and p . The system is therefore linear and non-autonomous, with time dependence entering through $q_{ij}(t)$ along the covariate path and through $\sigma_i^X(t)$ via (9).

Order of the drift perturbation.. Write $u_i = u_i^{(0)} + a u_i^{(1)} + O(a^2)$ for a perturbation of size a . The u block gains the forcing $a_i p_i$, so $u^{(1)}$ is $O(1)$ and u is first order in a . The w block gains $2a_i u_i$, a product of two quantities each first order in a after subtracting the baseline, so the variance response is $O(a^2)$. This is the analytic counterpart of the log-log slope of 2.0000 reported in Section 5.4.

We record one caveat against over-reading the result. The suppression concerns the variance channel and hence the leading effect on option value. For a non-at-the-money contract under a centred conditional-Gaussian perturbation, the first derivative of the call value with respect to a regime-dependent drift need not vanish identically, since the perturbation also reweights the regime mixture entering the terminal payoff. The numerical experiments in Section 5.4 bound the total effect at plausible magnitudes; they do not establish a general second-order proposition, and we do not claim one.

Appendix B. Transition reconstruction and label reconciliation

Label orientation.. The estimation output orders regimes by the raw fitted index, in which state 0 carries the higher diffusion coefficient and occupies roughly 67% of the sample with a mean duration of 7.56 hours, while state 1 carries the lower coefficient and a mean duration of 3.76 hours. Our convention is the opposite, with index 0 denoting the calm regime. Adopting it requires swapping the volatilities and, because the logistic coefficients are indexed by the same labels, swapping the transition coefficients crosswise: the production γ_{01} , governing calm-to-stress transitions, is the raw γ_{10} , and conversely. Failing to apply the swap to both objects would produce a model in which volatilities and transition dynamics refer to different states, which is a silent error rather than one that any single diagnostic would reveal. We enforce the ordering with an invariant check requiring $\sigma_0^y < \sigma_1^y$ at load time.

The orientation is worth a remark on its own terms. A high-volatility state that is both dominant and longer-lived inverts the usual reading in which stress is rare and brief. Whether this reflects genuine market structure at hourly frequency, or the labelling of a transform-space variance decomposition that does not correspond to economic stress, we cannot determine from the available artefacts, and we use the terms calm and stress throughout as labels for variance states rather than as economic claims.

Intercept reconstruction.. Slope coefficients and average marginal effects are supplied by the estimation output, but intercepts are not. The mean departure rate from state i over the covariate distribution must match the reported mean duration d_i , which gives

$$\mathbb{E}_z[\Lambda(\alpha_{01} + \gamma_{01}z)] = \frac{1}{d_{\text{calm}}}, \quad \mathbb{E}_z[\Lambda(\alpha_{10} + \gamma_{10}z)] = \frac{1}{d_{\text{stress}}}, \quad (\text{B.1})$$

for α_{01} and α_{10} separately, taking the expectation over the full standardised covariate sample and using a monotone bracketing root finder. The target rates are 0.2660 and 0.1323 per hour. The resulting pair reproduces the reported stationary stress share of 0.6746 to within 1.0% under an ergodic approximation and 4.5% under simulation along the realised covariate path.

This is moment matching, not estimation. The reconstructed intercepts have no sampling distribution, no standard errors and no likelihood interpretation, and any inference conditioned on their precise values would be unfounded. They are reported as Derived in Table 2 for that reason.

Appendix C. Numerical implementation and verification

Moment integration.. The system (12) is advanced with classical fourth-order Runge–Kutta and internal sub-stepping capped at 0.25 hours, with coefficient paths interpolated linearly within each step. The cap is necessary rather than conservative: on hourly steps without sub-stepping the second-moment trajectory exhibited visible instability at the calibrated switching intensities, which propagated into terminal dispersion.

Spatial discretisation.. The coupled system (14) is solved on a uniform grid of 1,201 nodes in the residual coordinate, with the domain set wide enough that the terminal density is numerically negligible at the boundaries. The regime coupling term is treated within the same implicit step as the diffusion, so the two value functions advance together.

Verification.. Three checks are run as automated regressions rather than as one-off diagnostics, so that any future change to the code that breaks them fails visibly.

1. *Put–call parity*: $\max|C - P - e^{-rT}(F(T) - K)| = 1.0 \times 10^{-6}$ TRY/MWh across the 66-point strike–maturity grid.
2. *Centring identity*: $\max_t |\mathbb{E}^\mathbb{Q}[P_t] - F(t)| < 10^{-8}$ TRY/MWh under drift perturbations including asymmetric and mixed-sign pairs, with a companion assertion that the uncentred residual mean is meaningfully non-zero so that the test cannot pass vacuously.
3. *Independent resimulation*: 20,000 Monte Carlo paths at a 0.05-hour step with an exact conditional Ornstein–Uhlenbeck update on the sampled regime path, giving 169.18 TRY/MWh against 166.75 TRY/MWh from finite differences at the reference contract.

We are explicit about what these checks do and do not establish. Parity and the centring identity test internal consistency of the implementation. The Monte Carlo comparison tests that two structurally different numerical routes to the same expectation agree. None of them provides evidence that the parameters describe the market, and the realised evaluation of Section 7.4 is the only exercise in this paper that bears on that question.

Reproducibility.. All tables and figures in this paper are generated from a versioned snapshot of the code and inputs. The full automated test suite comprises 207 tests covering the invariants above, the curve construction, the

parameter loader, provenance detection and the regression baselines reported here.

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